NETWORK / SOC ENGINEER L1 · INTERVIEW & FIELD GUIDE · PART 2 OF 2

Switching, Routing, TCP/IP & Firewall Troubleshooting

Theory · Configuration · Packet flow · Wireshark captures explained · Scenario questions · Customer-network troubleshooting

OSI & TCP/IP | ARP / ICMP | TCP / UDP | DNS / DHCP | Subnetting / NAT | VLAN / STP / EtherChannel | OSPF / BGP / HSRP | Palo Alto / IPSec / GlobalProtect | Wireshark

ABOUT THE WIRESHARK SCREENSHOTS

The capture images in this guide are Wireshark-style renderings of realistic example packets (field names, values and colours follow what Wireshark shows). They are for learning to read captures. They are not live screenshots from a real network, so IPs and MACs are lab examples. Open the same traffic in your own lab (GNS3 / EVE-NG / Packet Tracer + Wireshark) and compare.

Contents: Part 2 of 2

9	Firewall: Palo Alto flow, NAT, IPSec, GlobalProtect, Panorama
10	Wireshark field guide and filters
11	How routers communicate: hop-by-hop worked example
12	Troubleshooting a customer network (architecture + scenarios)
13	Rapid-fire interview Q&A
14	Last-day revision plan and cheat sheet

Part 1 contains: 1. OSI, TCP/IP and end-to-end packet flow; 2. Ethernet, MAC and ARP; 3. IP, ICMP, ping and traceroute; 4. TCP and UDP; 5. DNS and DHCP; 6. Subnetting and NAT/PAT; 7. Switching: VLAN, trunk, STP, EtherChannel, switch security; 8. Routing: static, OSPF, BGP, HSRP.

HOW TO USE THIS GUIDE

Order of study: chapters 1, 4, 2, 7 (OSI, TCP, ARP, switching) first, as you asked, then 8, 5, 3, 6, then Part 2 (firewall, Wireshark, router communication, customer troubleshooting, Q&A). Every chapter ends with scenario questions so you practise answering aloud.

9. Firewall: Palo Alto flow, NAT, IPSec, GlobalProtect, Panorama

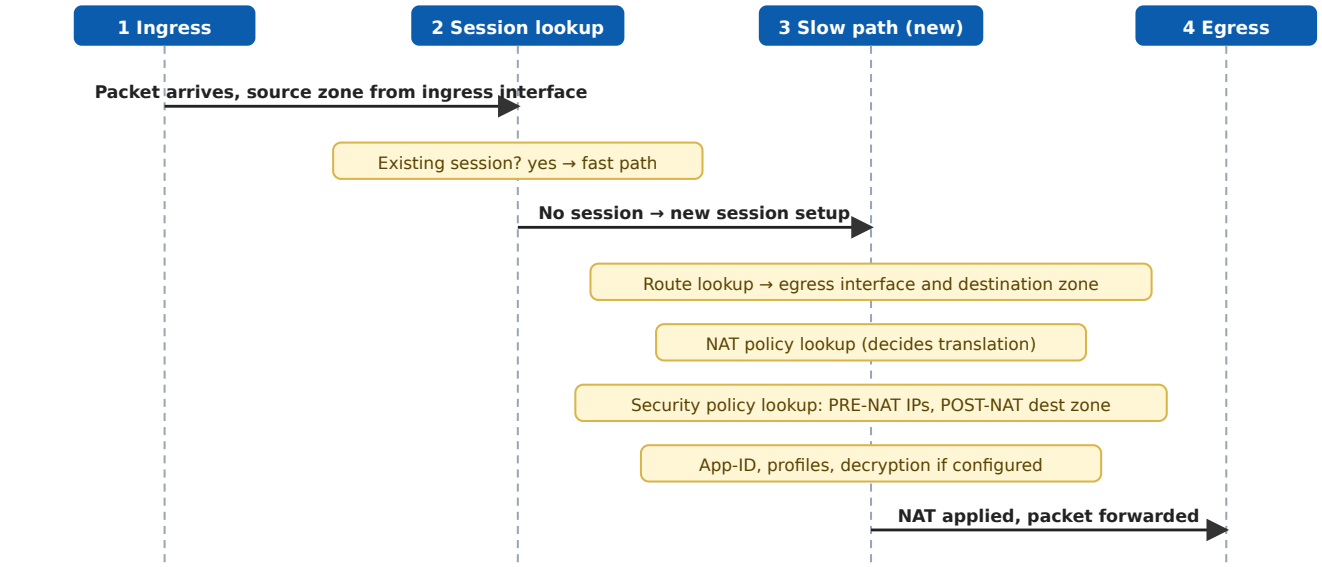
Your strongest area: be precise here

9.1 Core concepts

- **Zones** group interfaces (trust, untrust, dmz). Policy is zone-based. Interface types: Layer3, Layer2, Virtual Wire, Tap. **Virtual router** holds routes.
- **Security policy:** top-down, first match wins. Match on source/destination zone, address, user, **application (App-ID)**, service, URL category. Defaults: *intrazone-default = allow*, *interzone-default = deny* (not logged unless enabled).
- **Content-ID / profiles:** antivirus, anti-spyware, vulnerability, URL filtering, file blocking, WildFire. **User-ID** maps IP to user.
- **Session:** first packet goes through the full path (slow path); later packets use the session (fast path). Default timeouts: TCP 3600 s, UDP 30 s, ICMP 6 s.

First-packet processing order (simplified)

Policy uses the original addresses, but the destination zone is the one after NAT/route



9.2 Troubleshooting method: user cannot reach an application through the firewall

1. **Logs:** Monitor > Traffic. Filter by source/destination. Note rule, action, application, *session end reason* (policy-deny, aged-out, tcp-rst-from-client/server, threat, decrypt error). No log at all = traffic not arriving (routing/upstream) or implicit deny not logged.
2. **Zones/routing:** is traffic arriving on the expected interface/zone? `show routing route destination 10.1.1.50`, `test routing fib-lookup virtual-router default ip 10.1.1.50`.
3. **Policy:** `test security-policy-match from trust to untrust source 192.168.10.10 destination 10.1.1.50 destination-port 443 protocol 6`. Check rule order and shadowing.
4. **NAT:** `test nat-policy-match ...` and `show session id N` (shows c2s and s2c flows with NAT).
5. **Session:** `show session all filter source 192.168.10.10 destination 10.1.1.50`.
6. **Counters:** `show counter global filter delta yes severity drop` (e.g. flow_policy_deny, flow_tcp_non_syn, flow_fwd_l3_nortr, flow_session_notfound).
7. **Packet capture:** GUI Monitor > Packet Capture with stages receive / firewall / transmit / drop. Compare what arrived with what left.

```
> show session all filter source 192.168.10.10 destination 10.1.1.50 destination-port 443
ID      Application  State   Type Flag  Src[Sport]/Zone/Proto (translated IP[Port])     Dst[Dport]/Zone
1234    ssl          ACTIVE  FLOW  NS   192.168.10.10[51514]/trust/6 (203.0.113.2[40001])  10.1.1.50[443]/untrust
> show session id 1234
> test security-policy-match from trust to untrust source 192.168.10.10 destination 10.1.1.50 destination-port 443 protocol 6
> test nat-policy-match from trust to untrust source 192.168.10.10 destination 10.1.1.50 destination-port 443 protocol 6
> show counter global filter delta yes severity drop
> debug dataplane packet-diag set filter match source 192.168.10.10 destination 10.1.1.50
> debug dataplane packet-diag set filter on
> debug dataplane packet-diag set capture stage receive file rx.pcap
> debug dataplane packet-diag set capture on      ! reproduce, then: set capture off
> view-pcap filter-pcap rx.pcap
> debug dataplane packet-diag clear all
```

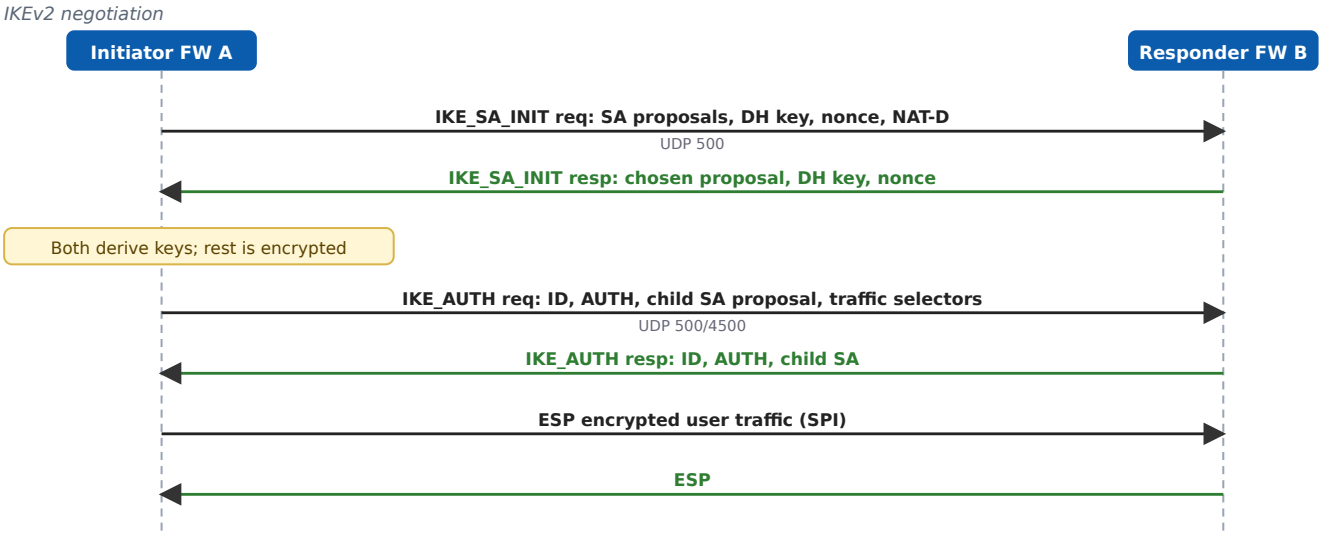
Log / symptom	Meaning	Action
Session end: policy-deny / action deny	Matched a deny or the interzone default	Fix rule order/match; check zone, App-ID, service
Application: incomplete	TCP handshake did not complete, or no data after it	Server not responding, return path, asymmetric routing
Application: insufficient-data / unknown-tcp	Not enough data to identify or custom app	Allow by service/port temporarily, create custom App-ID
Session end: aged-out, no reply	Silent drop upstream or no return traffic	Capture on both sides, check routing and server
flow_tcp_non_syn drops	Mid-stream packet without a session: asymmetric path or session expired	Make routing symmetric, check HA, timeout
flow_fwd_l3_nortr	No route	Add route in the right virtual router
Works for IP, fails for FQDN	DNS or FQDN object not resolved	Check DNS proxy / object refresh
decrypt-cert-validation / decrypt-unsupport-param	SSL decryption issues	Add no-decrypt exception, trust the CA, check cipher

9.3 NAT on the firewall

- Source NAT (dynamic IP and port) for internet access, destination NAT for published servers, static NAT for one-to-one.
- NAT rules are evaluated top-down, separate from security policy. Zones in the NAT rule are original source zone and **original** destination zone.
- Published server needs: NAT rule (public to private) **and** security rule with **public (pre-NAT) destination IP**, destination zone = zone of the private server.
- Server's default route must point back through the firewall (otherwise asymmetric).

9.4 IPSec VPN

	Phase 1 (IKE SA)	Phase 2 (IPSec / Child SA)
Purpose	Authenticate peers, create secure control channel	Create tunnels that carry user traffic
Parameters	Auth (PSK/cert), encryption, hash/PRF, DH group, lifetime, peer ID	Encryption, authentication, PFS/DH group, lifetime, proxy IDs / traffic selectors
Ports/protocol	UDP 500; UDP 4500 if NAT-T	ESP (IP protocol 50) or ESP in UDP 4500
IKEv1	Main mode (6 msgs) or aggressive (3)	Quick mode (3 msgs)
IKEv2	IKE_SA_INIT + IKE_AUTH (4 msgs)	Child SA is created inside IKE_AUTH; CREATE_CHILD_SA for rekeys



Wireshark: IKEv2 and ESP

ike.pcapng - Wireshark

Apply a display filter ... `isakmp || esp || udp.port==500 || udp.port==4500`

No.	Time	Source	Destination	Protocol	Len	Info
1	0.000000	198.51.100.10	203.0.113.2	ISAKMP	450	IKE_SA_INIT MID=00 Initiator Request
2	0.031000	203.0.113.2	198.51.100.10	ISAKMP	410	IKE_SA_INIT MID=00 Responder Response
3	0.060000	198.51.100.10	203.0.113.2	ISAKMP	322	IKE_AUTH MID=01 Initiator Request
4	0.095000	203.0.113.2	198.51.100.10	ISAKMP	286	IKE_AUTH MID=01 Responder Response
5	1.000000	198.51.100.10	203.0.113.2	ESP	142	ESP (SPI=0xc3a1f0e2)
6	1.021000	203.0.113.2	198.51.100.10	ESP	142	ESP (SPI=0x1b9d44aa)
7	60.000000	198.51.100.10	203.0.113.2	ISAKMP	450	IKE_SA_INIT MID=00 Initiator Request
8	60.030000	203.0.113.2	198.51.100.10	ISAKMP	90	IKE_SA_INIT MID=00 Responder Response NOTIFY: NO_PROPOSAL_CHOSEN

```
▸ Internet Protocol Version 4, Src: 198.51.100.10, Dst: 203.0.113.2
▾ User Datagram Protocol, Src Port: 500, Dst Port: 500 (1)
▾ Internet Security Association and Key Management Protocol
      Initiator SPI: 8c4d2e1a9b7f6a31 (2)
      Responder SPI: 0000000000000000
      Next payload: Security Association (33)
      Version: 2.0
      Exchange type: IKE_SA_INIT (34) (3)
      Flags: 0x08 (Initiator, No higher version, Request) (4)
      Message ID: 0x00000000
      Length: 408
    ▾ Payload: Security Association (33) (5)
          Proposal: ENCR_AES_CBC (256), PRF_HMAC_SHA2_256, AUTH_HMAC_SHA2_256_128, DH group 14 (2048-bit MODP)
    ▸ Payload: Key Exchange (34)
    ▸ Payload: Nonce (40)
    ▾ Payload: Notify (41) - NAT_DETECTION_SOURCE_IP (16388) (6)
    ▸ Payload: Notify (41) - NAT_DETECTION_DESTINATION_IP (16389)
```

#	What it denotes
1	IKE on UDP 500. If a NAT device is detected in the path, later messages move to UDP 4500 (NAT-T) and ESP is wrapped in UDP.
2	Initiator SPI identifies the IKE SA. Responder SPI is zero in the first message and filled in by the reply.
3	Exchange types: IKE_SA_INIT 34, IKE_AUTH 35, CREATE_CHILD_SA 36, INFORMATIONAL 37.
4	Flag bits: Initiator (0x08), Response (0x20). Only requests with no reply = peer unreachable or UDP 500 blocked.
5	**Proposal** lists encryption, integrity/PRF, DH group. Must overlap with the peer's config. Packet 8: **NO_PROPOSAL_CHOSEN** (notify 14) = no common proposal in Phase 1.
6	NAT-Detection payloads: each side hashes IP/port. A mismatch tells both there is NAT, so they switch to NAT-T (4500). Packets 5-6 show ESP with a different SPI each direction; if you see ESP only one way, the return path is broken.

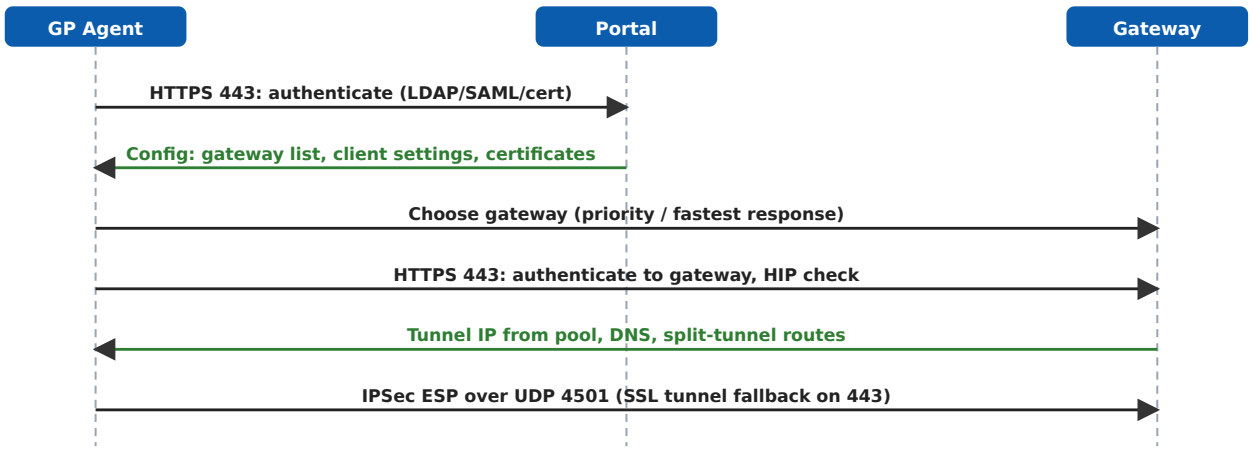
Figure 9.1: Tunnel bring-up (1-4), encrypted traffic (5-6), and a failed negotiation (7-8). ESP payload cannot be read without keys.

VPN problem	Check
Phase 1 down, no response from peer	Peer IP/route, UDP 500/4500 allowed upstream, interface/zone allows IKE (management profile or policy), peer configured. `show vpn ike-sa` , `less mp-log ikemgr.log` .
NO_PROPOSAL_CHOSEN / proposal mismatch	Align encryption, hash, DH group, IKE version between both ends (crypto profile).
Authentication failed	PSK mismatch, wrong local/peer ID, certificate issue.

Phase 1 up, Phase 2 fails	Proxy IDs / traffic selectors mismatch (Cisco ACL vs PAN proxy ID), PFS mismatch, wrong IPSec crypto profile.
Tunnel up, no traffic	Route to remote subnet via tunnel interface, security policy VPN zone, NAT exemption, overlapping subnets, tunnel monitor. `show vpn flow` (encap/decap counters): encap increasing but decap zero = remote not returning.
Tunnel drops periodically	Lifetime/rekey mismatch, dead peer detection, ISP flapping, NAT timeout (enable NAT keepalive).
Big transfers fail	MTU/MSS: adjust tunnel interface MTU / TCP MSS clamp.

9.5 GlobalProtect (SSL/IPSec remote access VPN)

GlobalProtect connection flow



GlobalProtect issue	Likely cause / check
Cannot connect to portal	DNS of portal FQDN, TCP 443 reachable (Wireshark SYN, no SYN-ACK), portal interface and policy, certificate trust (client does not trust the portal cert or chain).
Authentication fails	Auth profile (LDAP bind, SAML metadata/clock skew, RADIUS secret), user not in allowed list, certificate profile. Check system logs and `authd.log`.
Connected but no access to resources	Split-tunnel routes/access routes, security policy from GP zone to internal zone, NAT for tunnel pool, return route to pool on internal network.
Internal names do not resolve	DNS server/suffix not set in gateway config, DNS policy.
Frequently falls back to SSL / slow	UDP 4501 blocked on the user's network, so ESP-in-UDP fails, tunnel falls to SSL.
HIP check failing	HIP profile and object mismatch (patch/AV state); check HIP report in logs.
Check live users	`show global-protect-gateway current-user`; agent logs `PanGPS.log`, `PanGPA.log` on client.

9.6 Panorama and HA

- **Panorama** centrally manages firewalls. **Device groups** hold policies and objects (shared / pre-rules / post-rules). **Templates and template stacks** hold network and device settings (interfaces, zones, routing, profiles). **Commit** saves to Panorama, then **Push to Devices** applies it. Firewalls connect to Panorama on TCP 3978.
- Common issues: device shows *disconnected* (management reachability, TCP 3978, serial/auth key), *out of sync* (push needed or local override), commit failure from validation errors (missing zone/interface in template), template variable mismatch. Check `show panorama-status` (firewall) and `show devices connected` (Panorama).
- **High availability:** active/passive pair. HA1 control link and heartbeat, HA2 session/state sync, optional backup links. Failover triggers: link/path monitoring, heartbeat loss, manual suspend. Preemption optional. Verify `show high-availability state` and config sync status. Mismatched versions, licenses, or HA modes block sync.

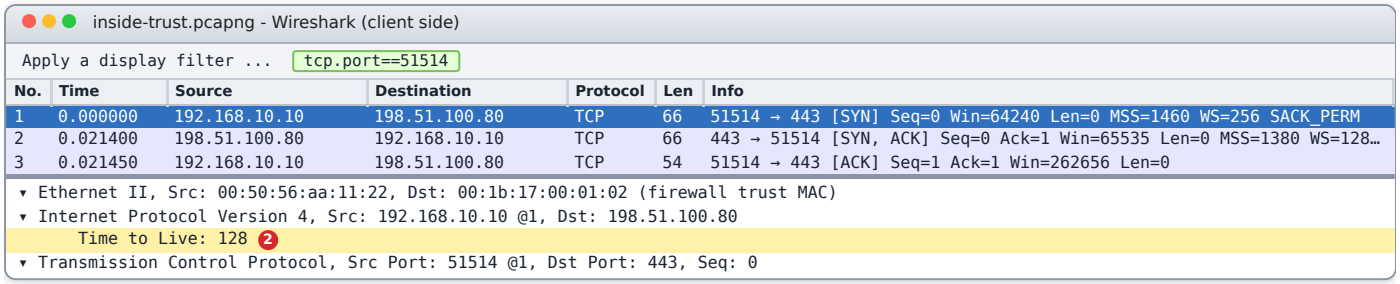
9.7 Deep dive: reading the traffic log

Traffic log field	Example	Why it matters
Source / Destination zone	trust → untrust	A wrong zone usually means traffic arrived on an unexpected interface or the route lookup chose another egress
Source / Destination	192.168.10.10 → 198.51.100.80	Original (pre-NAT) addresses are what policy matches

NAT Source / Destination	203.0.113.2:40001	Shows which translation was applied (blank = none)
Rule	Allow-Internet	The rule that matched. A deny rule or the interzone default explains most tickets
Application	ssl, web-browsing, incomplete	*incomplete* or *insufficient-data* means the handshake or data never finished
Session end reason	tcp-fin, aged-out, policy-deny, tcp-rst-from-server, threat	The first clue for 'connection drops'
Bytes sent / received	1.2 KB / 0 B	Zero received means the server never answered (path, return route, server)
Ingress / egress interface	ethernet1/2 → ethernet1/1	Confirms the path and helps spot asymmetry

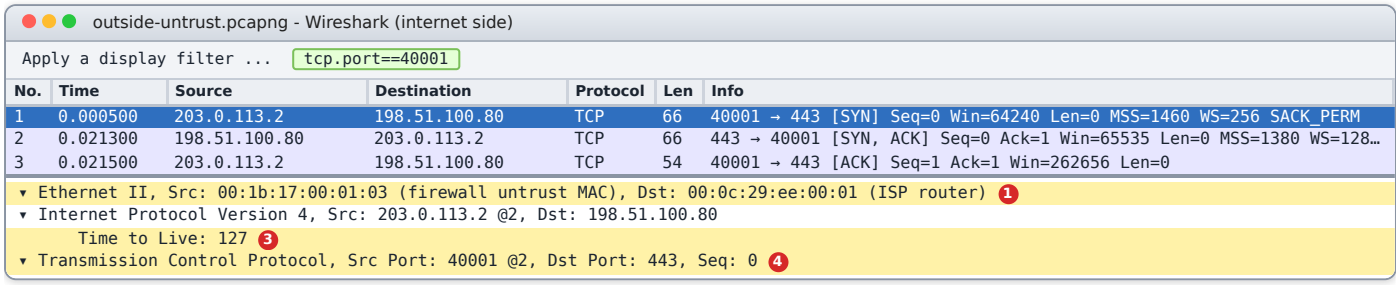
9.8 Wireshark: NAT seen from both sides of the firewall

Same connection captured on the inside (trust) and outside (untrust) interface. Only the source IP and port changed, plus the MAC addresses and TTL.



#	What it denotes
1	Original private source IP and port.
2	TTL is still the sender's initial value. Compare with the outside capture.

Figure 9.2: Inside interface. The server's MSS seen here is 1380 because something in the path clamped it.



#	What it denotes
1	New L2 header: the firewall's outside MAC to the next hop.
2	**Source NAT:** source IP and port are rewritten (PAT). The destination is untouched.
3	TTL dropped by 1 because the firewall (L3) routed the packet.
4	Sequence numbers are not changed by NAT. If they differ, a proxy or TCP-terminating device is involved.

Figure 9.3: Outside interface. If the inside capture shows traffic but the outside capture shows nothing, the firewall dropped it (check policy, routing, counters).

9.9 Wireshark: asymmetric routing and MTU problems

fw2-receive.pcapng - Wireshark (second firewall)

Apply a display filter ... `tcp.port==51600`

No.	Time	Source	Destination	Protocol	Len	Info
1	0.000000	192.168.10.10	10.1.1.50	TCP	54	51600 → 443 [ACK] Seq=1 Ack=1 Win=262656 Len=0
2	0.000900	192.168.10.10	10.1.1.50	TLSv1.2	571	Client Hello
3	0.001500	10.1.1.50	192.168.10.10	TCP	60	443 → 51600 [ACK] Seq=1 Ack=518 Win=64768 Len=0 [TCP ACKed unseen …

```
▾ Transmission Control Protocol, Src Port: 443, Dst Port: 51600, Seq: 1, Ack: 518
    ▾ [SEQ/ACK analysis]
        ▾ [TCP Analysis Flags]
              [Expert Info (Warning/Sequence): ACKed segment that wasn't captured (common at capture start)] (1)
```

#	What it denotes
1	This device never saw the SYN or SYN-ACK: the first packet it sees is a mid-stream ACK. A stateful firewall has no session, so it drops these (counter `flow_tcp_non_syn`). Cause: forward path through firewall 1, return path through firewall 2.

Figure 9.4: Asymmetric routing. Fix routing so both directions cross the same firewall, or use a design that syncs sessions.

pmtud.pcapng - Wireshark

Apply a display filter ... `icmp.type==3 || tcp.analysis.retransmission`

No.	Time	Source	Destination	Protocol	Len	Info
1	0.000000	192.168.10.10	10.1.1.50	TCP	15…	51700 → 443 [ACK] Seq=1 Ack=1 Win=262656 Len=1460
2	0.004100	10.0.0.1	192.168.10.10	ICMP	70	Destination unreachable (Fragmentation needed)
3	0.205000	192.168.10.10	10.1.1.50	TCP	15…	[TCP Retransmission] 51700 → 443 [ACK] Seq=1 Ack=1 Win=262656 Len=…
4	0.205800	10.0.0.1	192.168.10.10	ICMP	70	Destination unreachable (Fragmentation needed)

```
▾ Internet Control Message Protocol
      Type: 3 (Destination unreachable) (1)
      Code: 4 (Fragmentation needed) (2)
      MTU of next hop: 1400 (3)
▾ Internet Protocol Version 4 (original datagram, quoted)
      Flags: 0x40, Don't fragment (4)
```

#	What it denotes
1	Type 3 = unreachable.
2	Code 4 = fragmentation needed and DF set.
3	**The router tells the sender the maximum size it can forward.** A tunnel (IPSec/GRE) lowers the usable MTU.
4	Sender set DF, so the router cannot fragment. If ICMP is filtered somewhere, the sender never learns this and large transfers hang while small packets work (PMTUD black hole). Fix: allow ICMP type 3 code 4, lower MTU/MSS on the tunnel.

Figure 9.5: Path MTU discovery.

9.10 Wireshark: TLS to a GlobalProtect portal and certificate failure

gp-portal-tls.pcapng - Wireshark

Apply a display filter ... `tls`

No.	Time	Source	Destination	Protocol	Len	Info
1	0.000000	192.168.10.10	203.0.113.5	TLSv1.2	571	Client Hello
2	0.024000	203.0.113.5	192.168.10.10	TLSv1.2	14…	Server Hello, Certificate, Server Hello Done
3	0.026000	192.168.10.10	203.0.113.5	TLSv1.2	61	Alert (Level: Fatal, Description: Unknown CA)
4	0.026100	192.168.10.10	203.0.113.5	TCP	54	51801 → 443 [FIN, ACK] Seq=525 Ack=1401 Win=262656 Len=0

```
▾ Transport Layer Security
    ▾ TLSv1.2 Record Layer: Alert (Level: Fatal, Description: Unknown CA) (1)
          Content Type: Alert (21)
          Alert Message
              Level: Fatal (2) (2)
              Description: Unknown CA (48) (3)
```

#	What it denotes
1	The client closes the handshake itself.
2	Fatal = the connection is terminated.
3	**Unknown CA (48):** the client does not trust the issuer of the portal's certificate (or the intermediate CA is missing from the chain sent by the portal). Other codes: Certificate Expired (45), Bad Certificate (42), Certificate Unknown (46), Handshake Failure (40, no common cipher), Protocol Version (70).

Figure 9.6: Client rejects the portal certificate. Fix: install the issuing CA on clients or send the full chain from the portal, check expiry and the FQDN in the certificate (SAN).

9.11 Cisco to Palo Alto IPSec interoperability

Cisco (IKEv1 crypto map / ASA / ISR)	Palo Alto equivalent	Typical mismatch
crypto isakmp policy: encryption, hash, group, lifetime	IKE crypto profile (encryption, authentication, DH group, lifetime)	Different DH group or hash
crypto isakmp key + address	IKE gateway: pre-shared key, peer IP/ID	Different key, wrong peer ID type
transform-set (esp-aes esp-sha-hmac)	IPSec crypto profile (ESP encryption/authentication, DH group for PFS)	PFS enabled on one side only
crypto ACL (interesting traffic)	Proxy ID (local/remote subnet) on the tunnel	Subnets in ACL do not mirror the proxy ID; a single wide proxy ID vs several ACL lines
crypto map applied to interface	Tunnel interface + virtual router + zone + route	No route to remote subnet through tunnel.x

Tunnel status	What it tells you
IKE SA present, IPSec SA absent	Phase 1 fine; Phase 2 failing: proxy ID, PFS or crypto profile
Both present, encap counter rises, decap zero	Remote end is not returning traffic (route, policy, ACL, NAT)
Both present, both counters rise, app fails	Routing/policy/NAT/MTU on the inside
Tunnel flaps at lifetime	Rekey mismatch, DPD, NAT-T keepalive

INTERVIEW ANGLE

"Walk me through a VPN tunnel down ticket." Confirm which phase fails with *show vpn ike-sa* and logs (ikemgr.log, system log). Phase 1: reachability on UDP 500/4500, proposal, PSK, IDs. Phase 2: proxy IDs/selectors, PFS, crypto profile. Then data plane: routes, policy, NAT, encap/decap counters, MTU. Capture IKE if logs are unclear. Escalate with the logs, config of both sides, and timestamps.

10. Wireshark field guide and filters

Capturing, filtering and reading quickly

10.1 Capture workflow

1. Capture at the right place: client, server, firewall interface or SPAN port. Capture on both sides to localise loss.
2. Use **capture filters** (BPF) to limit what is recorded, e.g. `host 10.1.1.50 and port 443`. Use **display filters** to analyse afterwards.
3. Keep captures short, reproduce the issue once, note time of the failure.
4. Useful views: Statistics > Conversations, Protocol Hierarchy, I/O Graph, Analyze > Expert Information, right-click > Follow > TCP Stream.

Display filter	Use
`ip.addr == 10.1.1.50`, `ip.src ==`, `ip.dst ==`	Host traffic
`tcp.port == 443`, `udp.port == 53`	By port
`tcp.flags.syn==1 && tcp.flags.ack==0`	New connection attempts
`tcp.flags.reset==1`	Resets
`tcp.analysis.retransmission || tcp.analysis.duplicate_ack || tcp.analysis.zero_window`	TCP problems
`tcp.analysis.flags`	All TCP expert flags
`dns`, `dns.flags.rcode != 0`	DNS, DNS errors
`dhcp`, `bootp.option.dhcp == 1`	DHCP, Discover
`arp`, `arp.duplicate-address-detected`	ARP, duplicate IP
`icmp.type == 3 || icmp.type == 11`	ICMP errors
`stp`, `ospf`, `bgp`, `hsrp`, `isakmp`, `esp`	Protocols
`vlan.id == 20`	VLAN
`http.request || tls.handshake.type == 1`	HTTP requests / TLS Client Hello
`frame.time_delta > 1`	Gaps longer than 1 s
`tcp.stream eq 5`	One conversation

Capture filter (BPF)	Use
`host 10.1.1.50`	One host
`net 192.168.10.0/24`	Subnet
`port 53`, `tcp port 443`	Ports
`not arp and not broadcast`	Remove noise
`udp port 500 or udp port 4500`	IPSec negotiation

10.2 Reading a capture: 5-step approach

1. **Protocol Hierarchy / Conversations**: who talks most, anything unexpected.
2. **Expert Information**: retransmissions, resets, malformed packets, duplicate IPs.
3. Pick the failing conversation (`tcp.stream`) and read the handshake.
4. Check **time deltas** (`tcp.time_delta`, `dns.time`) to separate network delay from server delay.
5. Compare client-side and server-side captures: where does the packet disappear or change (NAT, TTL, MTU, flags)?

Colour in Wireshark (default)	Typically
Light purple	TCP
Light blue	UDP / DNS / DHCP
Pink	ICMP
Pale yellow/peach	ARP / routing protocols
Green	HTTP
Black with red text	TCP problems (retransmission, dup ACK, out-of-order)

Red with white text	TCP RST

10.3 Wireshark: slow server (application latency)

slow-server.pcapng - Wireshark

Apply a display filter ... `http`

No.	Time	Source	Destination	Protocol	Len	Info
1	0.000000	192.168.10.10	10.1.1.50	HTTP	412	GET /report?id=881 HTTP/1.1
2	0.000700	10.1.1.50	192.168.10.10	TCP	54	80 → 51900 [ACK] Seq=1 Ack=359 Win=64128 Len=0
3	4.213000	10.1.1.50	192.168.10.10	HTTP	15…	HTTP/1.1 200 OK (text/html)

```
▸ Frame 3: 1514 bytes on wire
▸ Transmission Control Protocol, Src Port: 80, Dst Port: 51900
▾ Hypertext Transfer Protocol
      HTTP/1.1 200 OK\r\n (1)
      [Time since request: 4.212300000 seconds] (2)
      [Request in frame: 1]
```

#	What it denotes
1	Status line: 2xx success, 3xx redirect, 4xx client error (401 auth, 403 forbidden, 404 not found), 5xx server error (500, 502 bad gateway, 503 unavailable, 504 gateway timeout).
2	The server's TCP stack ACKed in 0.7 ms (packet 2), so the network is fine. The 4.2 s gap is **server/application processing time**. Evidence for the application team.

Figure 10.1: Network vs application delay. Add the column `http.time` or `tcp.time_delta` to find slow responses quickly.

10.4 Wireshark: TLS handshake (what you can and cannot see)

tls13.pcapng - Wireshark

Apply a display filter ... `tls.handshake.type == 1`

No.	Time	Source	Destination	Protocol	Len	Info
1	0.000000	192.168.10.10	10.1.1.50	TLSv1.3	571	Client Hello (SNI=app.corp.local)
2	0.003200	10.1.1.50	192.168.10.10	TLSv1.3	14…	Server Hello, Change Cipher Spec, Application Data
3	0.005000	192.168.10.10	10.1.1.50	TLSv1.3	118	Change Cipher Spec, Application Data

```
▾ Transport Layer Security
    ▾ TLSv1.3 Record Layer: Handshake Protocol: Client Hello
          Handshake Type: Client Hello (1) (1)
          Version: TLS 1.2 (0x0303)  (legacy field)
        ▸ Cipher Suites (16 suites) (2)
        ▾ Extension: server_name (len=15) (3)
              Server Name: app.corp.local
        ▾ Extension: supported_versions (4)
              Supported Version: TLS 1.3 (0x0304)
          Extension: application_layer_protocol_negotiation (h2, http/1.1)
```

#	What it denotes
1	First message of the handshake.
2	Ciphers the client offers. No overlap with the server = Handshake Failure alert.
3	**SNI** shows the requested hostname in clear text, useful for identifying destinations even on encrypted traffic (and what URL filtering/decryption decisions use).
4	Supported versions: the real version negotiation in TLS 1.3. In TLS 1.3 the certificate is encrypted, so you will not see it in a capture (TLS 1.2 shows it).

Figure 10.2: TLS Client Hello.

10.5 Practical capture commands

```
# Linux server / host
tcpdump -i eth0 -nn -s0 -w /tmp/cap.pcap host 10.1.1.50 and port 443
tcpdump -i eth0 -nn arp or icmp
# Windows: Wireshark + Npcap, choose the right adapter, start capture before reproducing the problem.

# Cisco Catalyst embedded packet capture (syntax varies by platform/version)
monitor capture CAP interface GigabitEthernet1/0/5 both
monitor capture CAP match any
monitor capture CAP buffer size 10
monitor capture CAP start
monitor capture CAP stop
monitor capture CAP export flash:cap.pcap

# Switch SPAN (mirror) to a laptop running Wireshark
monitor session 1 source interface g1/0/5 both
monitor session 1 destination interface g1/0/10

# Palo Alto: GUI Monitor > Packet Capture (filter, then stage: receive / firewall / transmit / drop)
```

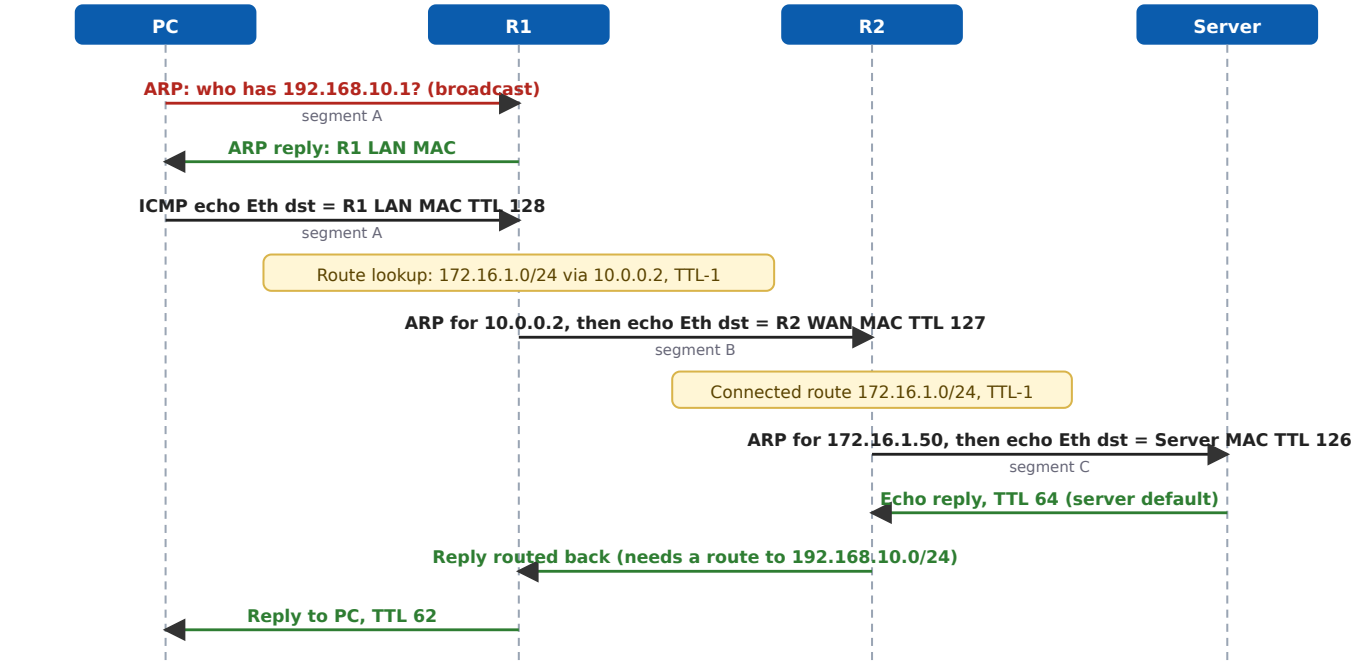
Wireshark skill	How
Add useful columns	Right-click field → Apply as Column: `tcp.stream`, `tcp.time_delta`, `ip.ttl`, `tcp.window_size`
See real sequence numbers	Edit → Preferences → Protocols → TCP → untick relative sequence numbers
Time since previous packet	View → Time Display Format → Seconds Since Previous Displayed Packet
Find where delay is	Filter `frame.time_delta_displayed > 1`
Reassemble a conversation	Right-click → Follow → TCP Stream (unencrypted payloads only)
Decode non-standard port	Analyze → Decode As
Throughput / loss over time	Statistics → I/O Graph; TCP Stream Graphs → Time Sequence (tcptrace)
Latency summary	Statistics → Conversations; Statistics → Service Response Time

11. How routers communicate: hop-by-hop worked example

What really changes in a packet as it crosses the network

Topology: PC `192.168.10.10` → R1 (LAN `192.168.10.1`, WAN `10.0.0.1`) → R2 (WAN `10.0.0.2`, LAN `172.16.1.1`) → Server `172.16.1.50`. The PC pings the server for the first time, so every ARP cache is empty.

First ping across two routers (each segment has its own ARP)



Segment	Eth src	Eth dst	IP src	IP dst	TTL
A: PC → R1	PC MAC	R1 LAN MAC	192.168.10.10	172.16.1.50	128
B: R1 → R2	R1 WAN MAC	R2 WAN MAC	192.168.10.10	172.16.1.50	127
C: R2 → Server	R2 LAN MAC	Server MAC	192.168.10.10	172.16.1.50	126

Segment A (PC NIC).pcapng - Wireshark

Apply a display filter ... icmp

No.	Time	Source	Destination	Protocol	Len	Info
1	0.000000	192.168.10.10	172.16.1.50	ICMP	74	Echo (ping) request id=0x0001, seq=1/256, ttl=128 (reply in 2)
2	0.001900	172.16.1.50	192.168.10.10	ICMP	74	Echo (ping) reply id=0x0001, seq=1/256, ttl=62

```
▾ Ethernet II, Src: 00:50:56:aa:11:22, Dst: 00:0c:29:cc:33:44 (1)
▾ Internet Protocol Version 4, Src: 192.168.10.10, Dst: 172.16.1.50 (2)
      Time to Live: 128 (3)
▸ Internet Control Message Protocol
```

#	What it denotes
1	Different MAC pair on every segment: L2 is rewritten by each router.
2	IP addresses are identical on all three captures.
3	TTL = 128: decreases by 1 per router. Capture at both ends to prove a router in the path is routing, not bridging.

Figure 11.1: Segment A (PC NIC).

Segment B (R1-R2 link).pcapng - Wireshark

Apply a display filter ... icmp

No.	Time	Source	Destination	Protocol	Len	Info
1	0.000000	192.168.10.10	172.16.1.50	ICMP	74	Echo (ping) request id=0x0001, seq=1/256, ttl=127 (reply in 2)
2	0.001900	172.16.1.50	192.168.10.10	ICMP	74	Echo (ping) reply id=0x0001, seq=1/256, ttl=63

▾ Ethernet II, Src: 00:0c:29:cc:33:55, Dst: 00:0c:29:dd:44:11 (1)
▾ Internet Protocol Version 4, Src: 192.168.10.10, Dst: 172.16.1.50 (2)
Time to Live: 127 (3)
▸ Internet Control Message Protocol

#	What it denotes
1	MACs on this segment.
2	Same IPs.
3	TTL 127.

Figure 11.2: Segment B (R1-R2 link).

Segment C (server NIC).pcapng - Wireshark

Apply a display filter ... icmp

No.	Time	Source	Destination	Protocol	Len	Info
1	0.000000	192.168.10.10	172.16.1.50	ICMP	74	Echo (ping) request id=0x0001, seq=1/256, ttl=126 (reply in 2)
2	0.001900	172.16.1.50	192.168.10.10	ICMP	74	Echo (ping) reply id=0x0001, seq=1/256, ttl=64

▾ Ethernet II, Src: 00:0c:29:dd:44:22, Dst: 00:50:56:bb:55:66 (1)
▾ Internet Protocol Version 4, Src: 192.168.10.10, Dst: 172.16.1.50 (2)
Time to Live: 126 (3)
▸ Internet Control Message Protocol

#	What it denotes
1	MACs on this segment.
2	Same IPs.
3	TTL 126.

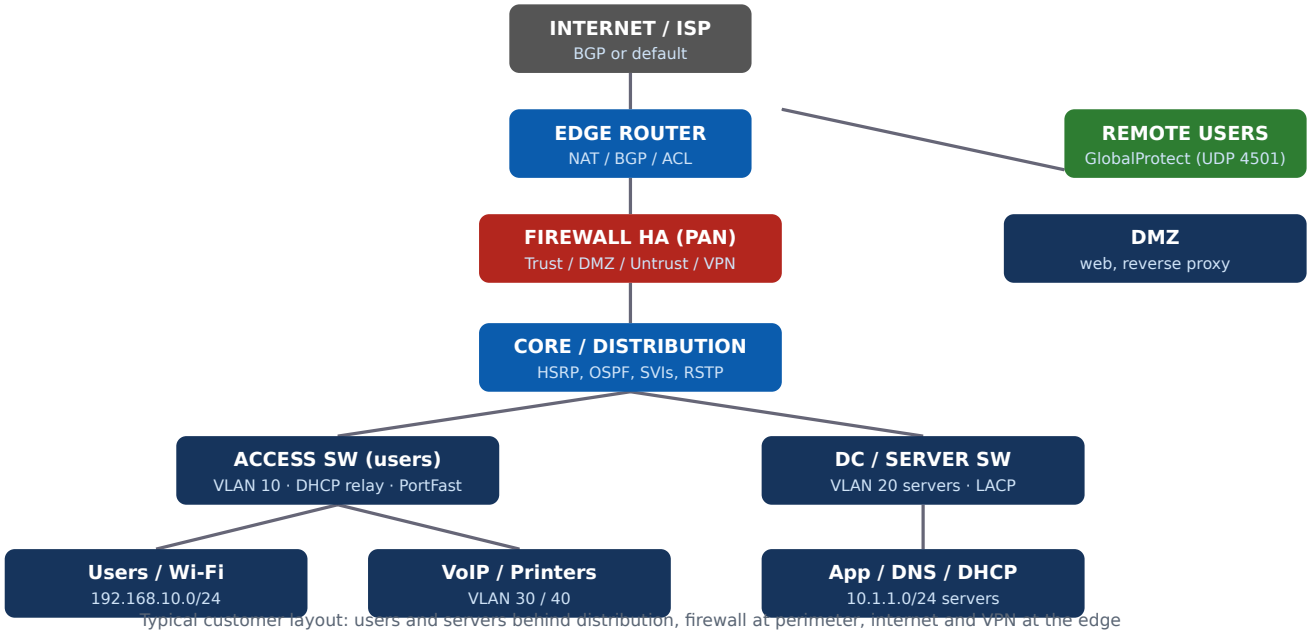
Figure 11.3: Segment C (server NIC).

If this fails...	Symptom	Cause
ARP for the gateway	Incomplete ARP entry on the PC	VLAN/port/gateway down
R1 has no route	ICMP net unreachable from R1 (3/0)	Missing route or routing protocol down
R1 cannot ARP the next hop	Host unreachable (3/1) or timeout	L2/VLAN problem on the WAN segment
R2 has no return route	Echo reaches the server, reply never returns	Missing route to 192.168.10.0/24 (both directions must be routed)
TTL reaches 0	Time exceeded (11/0) from the router in the loop	Routing loop
Firewall in path	Timeout with no ICMP error	Policy or asymmetry

INTERVIEW ANGLE
"What changes and what stays the same as a packet crosses routers?" MACs change every hop, TTL drops by one, the IP header checksum is recomputed. IPs and ports stay the same unless NAT is applied. The payload is unchanged.

12. Troubleshooting a customer network

Architecture-first method with worked scenarios



Typical customer layout: users and servers behind distribution, firewall at perimeter, internet and VPN at the edge

Reference architecture used in the scenarios below. Always ask the customer for a diagram, device list, IP plan and what changed recently.

12.1 Universal approach

1. **Define the problem:** who, what, since when, how many users, what changed? Get source/destination IP, port, application and the exact error message.
2. **Scope:** one user, one VLAN, one site, everyone? One destination or all? This points to access switch, distribution/gateway, firewall, or WAN/ISP.
3. **Follow the path** on the diagram, layer by layer: link → VLAN → gateway → routing → firewall/NAT → server → application.
4. **Test both directions** and from both ends (extended ping with source, traceroute, TCP port test).
5. **Collect evidence** (logs, counters, capture, config diff) and change one thing at a time.
6. **Document and escalate** with timeline, scope, tests run, evidence and impact.

Scope	Think first
One user	NIC/cable, VLAN, IP/DHCP, port security, local firewall, user policy
One VLAN / floor	Access switch, uplink trunk, STP, gateway SVI/HSRP, DHCP relay
All users to the internet	Firewall, NAT, edge router, ISP link, DNS, default route
One app only	Firewall policy/App-ID, server, DNS, load balancer, certificate
Remote users only	GlobalProtect/VPN, portal certificate, authentication, pool routing
Intermittent	Loop/flap, duplicate IP, HA failover, link errors, asymmetric path, session timeout

12.2 Questions to ask the customer first

- What exactly fails (application, URL, IP, port)? Exact error text and screenshot.
- When did it start? What changed (config, firmware, new VLAN, ISP, firewall policy, certificate renewal)?
- How many users and which location/VLAN? Is a colleague on the same floor affected?
- Does it fail by IP and by name? On wired and wireless? On VPN only?
- Is there a network diagram, IP plan, device list and monitoring access?

12.3 Worked scenarios

S1 One user has no network access

Symptom: PC shows 169.254.x.x, link light is on.

First hypotheses: Wrong VLAN, port err-disabled, DHCP not reaching the server.

#	Do this	What you see	What it means
1	`ipconfig /all` on the PC	IP 169.254.77.12, DHCP enabled	No Offer received (APIPA)
2	`show interfaces status` on access switch	Gi1/0/12 connected, vlan 10	Port is up and in the expected VLAN
3	`show mac address-table interface g1/0/12`	MAC present in VLAN 10	L2 is fine; problem is DHCP
4	Capture on the PC	Repeated DHCP Discover, no Offer	Server not answering or relay missing
5	`show run interface vlan 10` on gateway	No *ip helper-address*	Discover never leaves the VLAN
6	Check server scope list	No scope for 192.168.10.0/24 (new VLAN)	Server cannot choose a scope

ROOT CAUSE
VLAN 10 gateway lacks *ip helper-address* and the DHCP server has no scope for the subnet.

FIX AND VERIFICATION
Add the helper address and scope. On the PC run *ipconfig /renew*; verify Offer/ACK in a capture and *show ip dhcp snooping binding*.
Escalate if: the DHCP server is managed by another team and the scope exists; provide the Discover capture and relay config.

S2 Whole floor offline

Symptom: All users on floor 3 lost network at 10:05.

First hypotheses: Access switch down, uplink or trunk problem, STP change.

#	Do this	What you see	What it means
1	Ping the access switch management IP	Timeout	Switch or its uplink is unreachable
2	On distribution: `show interfaces status | include Gi1/0/24`	notconnect	Link down toward floor 3
3	`show logging | include LINK|UPDOWN|LINEPROTO`	Gi1/0/24 changed state to down at 10:05	Physical event, not config
4	`show interfaces transceiver detail`	Rx power very low / no light	Fibre or SFP fault
5	Ask onsite to check patching, swap SFP/patch lead	Link comes up after patch lead swap	Faulty patch cord

ROOT CAUSE
Damaged fibre patch lead on the uplink.

FIX AND VERIFICATION
Replace cable, confirm trunk, STP role and VLANs restored (*show interfaces trunk*, *show spanning-tree vlan 10*), users get DHCP again.
Escalate if: no light after cable and SFP swap (possible switch hardware or cabling plant fault); raise a hardware RMA.

S3 Users can ping the gateway but have no internet

Symptom: Intranet works. Websites fail for all users.

First hypotheses: Firewall/NAT, edge router, ISP, DNS, default route.

#	Do this	What you see	What it means
1	`ping 8.8.8.8` from a PC	Timeout	Not only a DNS issue
2	`traceroute -n 8.8.8.8`	Hop 1 gateway, hop 2 firewall, hop 3 no reply	Stops after the firewall
3	Firewall traffic log filter source 192.168.10.10	Entries allowed, bytes received = 0	Policy passes; no return traffic
4	`show routing route destination 8.8.8.8` on firewall	Default route via 203.0.113.1	Route exists

5	Ping 203.0.113.1 from the firewall (source = untrust IP)	No reply	ISP/next hop unreachable
6	Check untrust interface and edge router	Link up but ISP side down (BGP session Idle / provider outage)	Upstream failure

ROOT CAUSE
ISP link or BGP session down; firewall and policy are healthy.

FIX AND VERIFICATION
Open ticket with ISP with circuit ID and evidence; fail over to the backup link if configured (floating static / second BGP neighbour); verify after recovery.
Escalate if: provider confirms line is up (check VLAN/IP/ARP on the handoff).

S4 Internet works, an internal app does not

Symptom: Users cannot open https://app.corp.local (10.1.1.50).

First hypotheses: DNS, firewall policy, server, service port.

#	Do this	What you see	What it means
1	`nslookup app.corp.local`	Resolves to 10.1.1.50	DNS fine
2	`ping 10.1.1.50`	Replies	L3 path works (ICMP allowed)
3	`Test-NetConnection 10.1.1.50 -Port 443`	TcpTestSucceeded: False	Port closed or filtered
4	Capture on the client	SYN retransmissions, no response	Drop in the path
5	Firewall log for dst 10.1.1.50:443	Deny, rule interzone-default, new zone USERS-31	No rule for the new VLAN's zone

ROOT CAUSE
A newly created VLAN/zone has no security rule to the server zone.

FIX AND VERIFICATION
Add the rule or add the zone to the existing rule, commit, retest the TCP test, confirm hit count and log entries as allow.
Escalate if: the rule exists and the log shows allow but the server still does not answer (server team).

S5 Application is slow

Symptom: Users say 'the network is slow' for one web application.

First hypotheses: Network delay or loss, server latency, DNS, firewall decryption/CPU.

#	Do this	What you see	What it means
1	`ping -n 50 10.1.1.50`	0% loss, 2 ms	Network path is healthy
2	`nslookup` time / capture DNS	Responses in 3 ms	DNS fine
3	Capture: handshake and request	SYN-ACK in 1 ms; GET to response 4.2 s	Delay is on the server (see Figure 10.1)
4	Interface counters on path	No errors/drops	L1/L2 clean
5	Firewall CPU/session table	Normal	Not the firewall

ROOT CAUSE
Slow application/server response (for example a database query), not the network.

FIX AND VERIFICATION
Hand the capture and timing evidence to the application/server team; keep monitoring and agree a retest.
Escalate if: evidence shows network loss/latency on a specific hop (then escalate to the owner of that hop).

S6 Site-to-site VPN down

Symptom: Branch cannot reach HQ servers since 03:00.

First hypotheses: Peer IP changed, proposal/PSK change, ISP outage, Phase 2 mismatch.

#	Do this	What you see	What it means

1	`show vpn ike-sa`	No IKE SA for the branch gateway	Phase 1 not established
2	Ping the branch public IP; capture UDP 500/4500	Initiator requests but no response	Peer unreachable or blocking IKE
3	System log / ikemgr log	'IKE phase-1 negotiation failed, timed out'	No response from peer
4	Ask branch or ISP: did the public IP change?	ISP assigned a new dynamic IP at 03:00	Gateway peer IP is stale

ROOT CAUSE
The branch public IP changed and HQ still points to the old peer address.

FIX AND VERIFICATION
Update the peer address (or use a static IP/FQDN peer ID), commit, *test vpn ike-sa*, confirm IPSec SA and traffic counters.
Escalate if: branch device unreachable and no change at ISP.

S7 GlobalProtect users cannot connect

Symptom: Agent shows 'Connecting...' then failure for all users.

First hypotheses: Portal certificate, authentication backend, portal interface/policy.

#	Do this	What you see	What it means
1	From the internet: browse https://vpn.corp.com	Certificate warning / expired	Certificate issue
2	Capture or browser details	Expired yesterday	Handshake alert: Certificate Expired (45)
3	Check portal/gateway SSL/TLS profile	Using the expired certificate	Certificate not renewed

ROOT CAUSE
Expired portal/gateway certificate.

FIX AND VERIFICATION
Install the renewed certificate, update the SSL/TLS service profile, commit, test from an external client, confirm users connect (*show global-protect-gateway current-user*).
Escalate if: certificate is valid but authentication fails (check LDAP/SAML/RADIUS and clock).

S8 Intermittent drops every few minutes

Symptom: Applications reset roughly every 5 minutes for some users.

First hypotheses: Duplicate IP, MAC flap, STP changes, HA flapping, link flaps.

#	Do this	What you see	What it means
1	`show logging | include MACFLAP|DUPLEX|STP`	MACFLAP: host flapping between Gi1/0/5 and Gi1/0/9	A loop or duplicate MAC
2	`show spanning-tree detail | inc ieee|occur|from`	Topology changes every few minutes from Gi1/0/9	A port is flapping/looping
3	Trace Gi1/0/9	Goes to an unmanaged desk switch	Possible patch loop
4	Disconnect that cable and watch logs	MACFLAP stops	Confirmed

ROOT CAUSE
A small unmanaged switch connected twice to the access switch (layer-2 loop).

FIX AND VERIFICATION
Remove the loop, enable BPDU Guard and storm-control on access ports, verify no new TCN/MACFLAP messages.

S9 Failure after an HA/HSRP failover

Symptom: After failover some sessions work, others fail.

First hypotheses: Session sync, asymmetric path, ARP/MAC update, missing config on the secondary.

#	Do this	What you see	What it means
1	Check HA/HSRP state on both devices	Both claim active (split-brain) or standby has stale config	Hello/heartbeat path broken or config drift
2	Compare route tables and policy hit counts	Secondary missing a static route	Config out of sync

3	Capture hellos (HSRP UDP 1985 / HA1)	No hellos from peer	Link or VLAN between devices

ROOT CAUSE
Configuration difference and a broken heartbeat link.

FIX AND VERIFICATION
Restore the heartbeat path, sync configuration, test controlled failover in a window.

S10 Published server unreachable from the internet

Symptom: Public web server not reachable from outside.

First hypotheses: DNS, NAT, security policy, server gateway.

#	Do this	What you see	What it means
1	Resolve public name and test port from outside	Resolves to 203.0.113.50; port 443 timeout	Check inside the firewall
2	Firewall log for dst 203.0.113.50	No entries	Traffic not arriving or no matching log
3	Check upstream routing to 203.0.113.50	ISP routes the /29 to the firewall	OK
4	Test NAT and security policy match for that IP	Security rule written with the private IP as destination	Policy must use the pre-NAT (public) destination IP

ROOT CAUSE
The security rule used the translated address instead of the original destination.

FIX AND VERIFICATION
Edit the rule to match 203.0.113.50 with the server's zone as destination zone, commit, test from outside, confirm session and server gateway returns via the firewall.

S11 OSPF or BGP neighbour down

Symptom: Branch router lost its routes after a change window.

First hypotheses: Parameter mismatch, interface/MTU change, filter.

#	Do this	What you see	What it means
1	`show ip ospf neighbor`	Neighbour in EXSTART	Adjacency stuck
2	`show interface g0/0 | include MTU` both ends	1500 vs 1400	MTU mismatch (tunnel or changed interface)
3	Capture DBD packets	Interface MTU field differs	Confirms

ROOT CAUSE
MTU mismatch after a change.

FIX AND VERIFICATION
Align MTU on both sides (or *ip ospf mtu-ignore* as a temporary measure), confirm FULL state and routes in *show ip route ospf*. For BGP, apply the same method with TCP 179 reachability, AS and neighbour IP.

S12 Broadcast storm

Symptom: Network unusable, switch CPU 100%, links at full load.

First hypotheses: Layer-2 loop from patching or a new switch.

#	Do this	What you see	What it means
1	`show processes cpu | exclude 0.00`	High CPU in ARP/STP/switching processes	Storm
2	`show interfaces | include broadcasts|line protocol`	Two ports with huge broadcast counts	Loop path
3	Shut one of the two ports	Utilisation falls	Found the loop
4	Check what is cabled to those ports	Same VLAN connected twice by a new switch	Cause identified

ROOT CAUSE

Redundant uplinks to an unmanaged or misconfigured switch without STP protection.

FIX AND VERIFICATION

Remove or configure the device, enable BPDU Guard, Root Guard and storm-control, confirm the root bridge and port roles, write the RCA.

12.4 Escalation note template

```
Ticket: INC-xxxxx   Priority: P2   Time detected: 14:05 IST
Impact: 40 users on floor 3 cannot reach ERP (10.1.1.50:443). Others unaffected.
Findings: gateway reachable; SYN sent, no SYN-ACK; firewall log shows deny by 'interzone-default'.
Tests done: ping/traceroute (path OK to firewall), test security-policy-match, client capture.
Recent change: VLAN 31 added today (no policy for zone USERS-31).
Action requested: add rule or confirm zone. Evidence attached (pcap, logs).
```

13. Rapid-fire interview Q&A

Read these out loud. If you cannot answer in 30 seconds, revisit the chapter

Networking basics

Question	Short answer (say it in 20-30 seconds)
Difference between hub, switch, router?	Hub repeats bits to all ports (one collision domain). Switch forwards frames by MAC (a collision domain per port, one broadcast domain per VLAN). Router forwards packets by IP between broadcast domains.
Broadcast vs collision domain?	Collision domain: where frames can collide (per switch port in full duplex it is effectively none). Broadcast domain: where a broadcast is received (a VLAN/subnet). Routers and VLANs split broadcast domains.
What is the default gateway?	Router interface a host uses to reach other subnets.
Unicast, multicast, broadcast?	One-to-one, one-to-group (224.0.0.0/4), one-to-all in the subnet (255.255.255.255 or subnet broadcast).
What is MTU and MSS?	MTU: largest L3 packet on a link (1500). MSS: largest TCP payload per segment (MTU minus IP and TCP headers = 1460).
What is a VLAN and why use it?	Logical broadcast domain on shared hardware. Segmentation, security, smaller broadcast domains.
What is CIDR?	Classless addressing with prefix length (/24) instead of classful boundaries; enables VLSM and route summarisation.
Half vs full duplex?	Half: one direction at a time (CSMA/CD). Full: send and receive simultaneously. Mismatch causes late collisions and CRC errors.

TCP/UDP, ports and Wireshark

Question	Short answer (say it in 20-30 seconds)
Explain the TCP 3-way handshake.	SYN, SYN-ACK, ACK. Establishes sequence numbers and options (MSS, window scale, SACK).
What is the difference between RST and FIN?	FIN: graceful close of one direction. RST: abort immediately (closed port, error, firewall reset).
What does the TCP window do?	Receiver tells the sender how many bytes it can accept: flow control. Zero window = receiver full.
What causes retransmissions?	Packet loss or delay beyond RTO, or 3 duplicate ACKs (fast retransmit).
Which protocols use UDP and why?	DNS, DHCP, SNMP, syslog, NTP, IKE, VoIP: low overhead, speed, or app-level reliability.
How do you find the problem when a site is slow?	Wireshark: handshake RTT, DNS time, time to first byte, retransmissions, window problems; server vs network time.
What is the difference between capture and display filter?	Capture filters (BPF) limit what is recorded; display filters narrow what you see in an existing capture. Different syntax.
How do you see only SYN packets?	`tcp.flags.syn==1 && tcp.flags.ack==0`

ARP, ICMP, DNS, DHCP

Question	Short answer (say it in 20-30 seconds)
What does ARP do and what type of message is the request?	Maps IP to MAC. Broadcast request, unicast reply.
What is gratuitous ARP?	Unsolicited ARP announcing own IP/MAC; used for duplicate detection and failover.
What if two hosts have the same IP?	Intermittent connectivity, ARP cache flips, Windows warning; find with duplicate-IP expert info and MAC table.
How does ping work?	ICMP echo request (type 8) and echo reply (type 0).
What does TTL exceeded mean?	Packet's TTL reached 0 at a router (type 11). Used by traceroute; repeated hops mean a loop.
DNS resolution steps?	Cache, resolver, root, TLD, authoritative; answer cached for TTL. UDP/53 (TCP for large).
What is NXDOMAIN?	The name does not exist (rcode 3).

DHCP process?	Discover, Offer, Request, ACK (DORA); lease renewal at 50%, rebind at 87.5%. Relay via ip helper-address.
Client has 169.254.x.x?	APIPA: no DHCP response. Check VLAN, helper, scope, snooping.

Switching

Question	Short answer (say it in 20-30 seconds)
Access vs trunk?	Access: one VLAN, untagged. Trunk: multiple VLANs with 802.1Q tags.
What is native VLAN?	The untagged VLAN on a trunk; must match both ends.
How does STP choose the root bridge?	Lowest bridge ID (priority, then VLAN/sys-id, then MAC).
Port roles in STP?	Root, designated, alternate (blocked), backup.
STP vs RSTP?	RSTP converges in seconds with proposal/agreement and edge ports; STP takes 30-50 s.
PortFast and BPDU Guard?	PortFast: edge port forwards immediately. BPDU Guard: err-disables the port if a BPDU arrives (prevents rogue switches/loops).
EtherChannel and its modes?	Bundles links. LACP active/passive, PAgP desirable/auto, or on. Members must match.
How do you detect a loop?	Broadcast storm, 100% utilization, MAC flaps in logs, STP TC changes, high CPU.
What is port security?	Limits/learns MACs per port; violation modes protect/restrict/shutdown.
What is DAI / DHCP snooping?	Snooping: trusted/untrusted ports, builds binding table. DAI uses it to validate ARP.
What is inter-VLAN routing?	Router-on-a-stick or L3 switch SVIs route between VLANs.
What is a CRC error?	Frame failed FCS: bad cable, duplex mismatch, interference, bad port/optic.
Why is a port err-disabled?	Violation: BPDU guard, port security, link flap, UDLD, etc. Fix cause then bounce port.

Routing

Question	Short answer (say it in 20-30 seconds)
Static vs dynamic routing?	Static: manual, predictable, no overhead, no automatic failover. Dynamic: protocols adapt to changes (OSPF, BGP).
What is administrative distance?	Trustworthiness of a route source; lower wins between different protocols (Connected 0, Static 1, eBGP 20, OSPF 110, iBGP 200).
Longest prefix match?	The most specific matching route is used before AD or metric.
OSPF neighbor states?	Down, Init, 2-Way, ExStart, Exchange, Loading, Full.
OSPF stuck in ExStart?	MTU mismatch (classic) or duplicate router ID.
Parameters that must match for OSPF adjacency?	Area ID, hello/dead timers, subnet/mask, authentication, stub flags, MTU (for full), unique router IDs.
What is DR/BDR?	Broadcast-segment election to reduce adjacencies; priority then router ID.
eBGP vs iBGP?	eBGP between different AS (AD 20, TTL 1), iBGP within an AS (AD 200, no re-advertising iBGP routes).
BGP stuck in Active?	TCP/179 not establishing: reachability, ACL, wrong neighbor/AS, source interface.
BGP best-path order?	Weight, local-pref, locally originated, AS-path, origin, MED, eBGP over iBGP, IGP metric, router ID.
HSRP active/standby and preempt?	Virtual IP shared; highest priority active; preempt allows the higher priority to retake the role.
What is a floating static route?	Backup static with a higher AD that activates only if the primary route disappears.

Firewall and VPN (Palo Alto)

Question	Short answer (say it in 20-30 seconds)
What is a zone?	Logical grouping of interfaces; policy is between zones.
How does the firewall process a new session?	Ingress, session lookup (none), route/zone, NAT policy lookup, security policy lookup, session creation, App-ID/Content-ID, NAT applied, egress.
Security policy uses pre-NAT or post-NAT address?	Pre-NAT IPs, post-NAT destination zone.
What is App-ID?	Identifies applications by signatures and behaviour regardless of port.

Difference between deny, drop, reset?	Drop silent; deny uses the app's default deny; reset sends RST.
Phase 1 vs Phase 2?	Phase 1 builds IKE SA (authentication, key exchange); Phase 2 builds IPSec SA (data tunnel, proxy IDs).
Common Phase 1 failures?	Proposal mismatch, PSK/ID mismatch, peer unreachable, UDP 500/4500 blocked, NAT-T.
Tunnel up but no traffic?	Routes, policy, NAT exemption, proxy ID, remote side, MTU; check encap/decap counters.
What is GlobalProtect?	Remote-access VPN: portal for config, gateway for tunnel, agent on client, HIP checks.
What is Panorama?	Central management: device groups (policy/objects), templates (network/device config), logging.
How do you verify traffic path on a PA?	Traffic log, `show session`, `test security-policy-match`, routing lookup, packet capture at four stages.
What is HA failover trigger?	Link/path monitor failure, heartbeat loss, device failure; sync config (HA1) and sessions (HA2).
Why can asymmetric routing break a firewall?	Return packet arrives without a session; a non-SYN first packet is dropped.

SOC / L1 operations (basic awareness)

Question	Short answer (say it in 20-30 seconds)
What does an L1 do with an alert?	Triage: validate, enrich (IP/domain reputation, asset, user), classify true/false positive, document, escalate with evidence per runbook.
True positive vs false positive?	True: real malicious/policy event. False: benign activity flagged.
What is a SIEM?	Collects and correlates logs from many sources, raises alerts and supports investigation. Honest line if asked about specific tools: say you have not used that tool hands-on yet and describe how you worked with firewall logs and threat logs.
Port scan in logs?	One source hitting many ports/hosts quickly with SYNs and few completions; verify scope, block/escalate.
Brute force?	Many failed logins from one source or against one account; check source reputation, lockouts, success after failures; escalate if success.
ITIL incident vs problem vs change?	Incident: restore service. Problem: root cause of recurring incidents. Change: controlled modification with approval and rollback.
How do you handle 24x7 shifts and handover?	Clear shift handover notes: open tickets, pending actions, changes in progress, risks.

14. Last-day revision plan and one-page cheat sheet

Use this the evening before and the morning of the interview

14.1 Suggested schedule

Block	Do	Chapters
1	Read OSI/TCP/ARP/ICMP aloud; draw the packet flow from memory	1, 2, 3, 4
2	Switching: VLAN, STP root election, EtherChannel; draw the STP triangle	7
3	Routing: OSPF states + causes, BGP states + causes, HSRP	8
4	Firewall + VPN troubleshooting method, commands; prepare your real case	9, 12
5	Wireshark filters and how you read a handshake/RST/retransmission	4, 10
6	Rapid-fire Q&A aloud; 3 customer scenarios aloud with the 5-step method	12, 13
7	Self-introduction, STAR story, questions to ask; sleep	14

14.2 Port and protocol cheat sheet

Port/proto	Service	Port/proto	Service
TCP 20/21	FTP data/control	UDP 161/162	SNMP / traps
TCP 22	SSH	TCP/UDP 389, 636	LDAP / LDAPS
TCP 23	Telnet	TCP 443	HTTPS
TCP 25	SMTP	TCP 445	SMB
TCP/UDP 53	DNS	UDP 514	Syslog
UDP 67/68	DHCP server/client	UDP 500, 4500	IKE, NAT-T
UDP 69	TFTP	IP proto 50 / 51	ESP / AH
TCP 80	HTTP	TCP 3389	RDP
UDP 123	NTP	TCP 179	BGP
IP proto 89	OSPF	IP proto 88	EIGRP
UDP 1985	HSRP	IP proto 112	VRRP
UDP 1812/1813	RADIUS auth/acct	TCP 49	TACACS+
UDP 4501	GlobalProtect IPSec	TCP 3978	Palo Alto device ↔ Panorama

14.3 Numbers to remember

Topic	Values
OSPF	Hello 10 / dead 40 (broadcast, p2p); multicast 224.0.0.5/.6; AD 110; area 0 mandatory
BGP	TCP 179; keepalive 60 / hold 180; eBGP AD 20, iBGP 200; eBGP TTL 1
STP	Hello 2, max age 20, forward delay 15; default priority 32768; root = lowest bridge ID
HSRP	Hello 3 / hold 10; default priority 100; v1 224.0.0.2, v2 224.0.0.102; UDP 1985
Cache and aging	MAC table 300 s; Cisco ARP 4 h; DHCP renew 50%, rebind 87.5%
TTL defaults	Windows 128, Linux/macOS 64, Cisco 255
MTU / MSS	1500 / 1460; 802.1Q adds 4 bytes; IPSec overhead roughly 50-70 bytes
Palo Alto timeouts (defaults)	TCP session 3600 s, UDP 30 s, ICMP 6 s
Private ranges	10/8, 172.16/12, 192.168/16; APIPA 169.254/16
TCP flags	SYN start, ACK ack, FIN close, RST abort, PSH push

14.4 Fast command reference

Task	Command
Cisco interface health	`show ip interface brief` · `show interfaces status` · `show interfaces g1/0/1`
VLAN/trunk/MAC	`show vlan brief` · `show interfaces trunk` · `show mac address-table`
STP/EtherChannel	`show spanning-tree vlan 10` · `show etherchannel summary`
Routing	`show ip route` · `show ip ospf neighbor` · `show ip bgp summary` · `show standby brief`
Host	`ipconfig /all` · `arp -a` · `nslookup` · `tracert -d` · `netstat -ano`
Palo Alto	`show session all filter …` · `test security-policy-match` · `show counter global filter delta yes severity drop` · `show vpn ike-sa` · `show high-availability state`

14.5 Your story: structure to fill in with your own facts

TELL ME ABOUT YOURSELF (60-90 S)
1) Who you are: Network Security Engineer, about 1.4 years in a TAC role supporting enterprise Palo Alto NGFW environments. **2)** What you did daily: troubleshooting Panorama, IPSec/SSL VPN, GlobalProtect cases; Wireshark and packet captures; writing RCA reports. **3)** Foundation: B.Tech CS; CCNA preparation, labs and a GitHub portfolio on routing/switching/security. **4)** Why this role: want hands-on LAN/DC network operations and 24x7 monitoring that builds on firewall strength toward Network Security Engineering. Keep it to the facts that are true for you; adjust wording.

STAR STORY SKELETON (USE ONE REAL CASE)
Situation: customer, environment, symptom (e.g. site-to-site VPN flapping / users cannot reach an app through the firewall). **Task:** your responsibility and priority/SLA. **Action:** what you checked in order (logs, session, routing, NAT, capture stage), the finding (e.g. proxy-ID mismatch, asymmetric path, MTU), how you proved it with evidence. **Result:** fix, time to resolve, RCA delivered, prevention. Practise it to 90 seconds.

If they ask about…	Honest, confident answer
Cisco/Aruba/Juniper experience	"My production experience is Palo Alto firewalls and the routing/switching behind them. I have studied Cisco switching/routing for CCNA and practised in labs, and I learn new platforms quickly."
SDN / SDA / SIEM tools you have not used	"I have not worked on that platform hands-on yet. I understand the concept (explain in one line) and I am comfortable with logs, alerts and escalation procedures from my TAC work."
24x7 rotational shifts	"Yes, comfortable. I understand the importance of clean handovers and ticket documentation."
A mistake you made	Choose a real small one, what you learned, and the process you now follow (e.g. verify scope and take config backup before change).

14.6 Questions you can ask them

- What does a typical shift look like, and how are incidents escalated from L1?
- Which monitoring/ticketing tools and network platforms will I use (and is training provided)?
- What is the environment: size of the LAN/DC, vendors, SDA/SDN components?
- How will I be evaluated in the first 90 days?
- What learning or certification path is available (CCNA/CCNP, vendor training)?

FINAL REMINDERS
Day of the interview: sleep, water, laptop/phone charged, stable internet, quiet room. Answer with structure ("First I check scope, then layer 1-2, then..."). If you do not know something, say what you would check, and ask for the chance to reason it out. Do not guess silently.